Enhancing Breast Cancer Detection Through Advanced Machine Learning Techniques

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Abstract

Breast cancer is a dangerous and common disease that affects women all over the world. It is impossible to exaggerate the importance of early identification because it significantly improves patient outcomes and survival rates. Modern machine learning methods have presented a viable way to increase the accuracy and efficacy of breast cancer detection in recent years. This study investigates sophisticated machine and deep learning techniques to combine image attributes from digitalized fine needle aspirate (FNA) pictures with the Multisurface Method-Tree (MSM-T) classification technique. This work uses the Wisconsin (diagnostic) dataset, which consists of 65% data of one class and 35% of another class. To rectify class imbalances, the study conducts a thorough exploratory data analysis and uses exacting preprocessing methods. We split the dataset into three-fold and augmented the train set data using synthetic minority oversampling techniques (SMOTE) to balance the imbalanced class. The shortcomings of conventional approaches for detecting breast cancer serve as a sobering reminder of the need for more precise and reliable diagnostic techniques. As a result, a range of sophisticated machine learning methods, such as K-nearest neighbours, support vector machines (SVMs), fully connected neural networks, and random forests, are carefully examined to determine how well they can identify breast cancer patterns in the dataset. Using multilayer ensemble approaches, we combine all the different algorithms, and the model shows remarkable predictive power and skillfully detects important features. In addition, an evaluation using the confusion matrix lens highlights the model's effectiveness by demonstrating lower rates of false negatives and higher recall rates. The confusion matrix values show three different folds, and one combines all three different folds with mean and standard deviation values. Employing this thorough investigation, the present study aims to augment the toolkit of techniques accessible for addressing breast cancer, aiming to optimise diagnostic precision and ameliorate patient consequences.

Keywords. *Breast Cancer; Machine Learning; Exploratory data analysis; Confusion Matrix; Recall*

1 Introduction

Breast cancer is one of the most common cancers found in females. It is a disease where unusual breast cells from the mammary glands and ducts grow out of control and form malignant growth, if left unchecked can become fatal. According to Naji et al. [4], SVM, RF, LR, DT, and KNN are used to detect breast cancer. This machine's accuracy was 97.2%. Sharma et al [5] used SVM, RF, and NB to detect breast cancer using the WBCD. RF, KNN, and NB accuracy was 94.7% 95.9% and 94.47%, respectively. Research by Sengar et al. [6] present breast cancer detection technique using LR and DT. The study found that LR and DT were $94.40\% and 95.10\%$ accurate, respectively. Agarap [7] advise employing gated recurrent unit (GRU)-SVM, LR, multi-layer perceptron (MLP), L1-NN, L2-NN, SoftMax regression, and SVM.

2 Related work

2.1 Biological approaches

There are different biological approaches for the prognosis or to detect the stage of the breast cancer. This include Blood tests, such as complete blood count and tests to show how well the kidneys and liver are working and also include mammograms, CT scan , ultrasound and biopsies.The biological approaches take significant amount of time so there is a need to utilise computer based diagnostic methods to identify cancer at an early stage.

2.2 Machine learning based approaches

This methodology uses algorithms to diagnose tumors faster, more accurately, and in less time. The different machine learning algortihms used here are Multisurface Method-Tree (MSM-T) Classification Technique,K-Nearest Neighbors (KNN) which is a machine learning algorithm for classification based on similarity measures,Support Vector Machines (SVMs) used for classification tasks, particularly effective in high-dimensional spaces,Random Forests that constructs multiple decision trees and aggregates their predictions,Synthetic Minority Oversampling Technique (SMOTE) used for dealing with class imbalance by generating synthetic examples of the minority class.The study additionally looks into ensemble approaches, which combine predictions from several algorithms (potentially including KNN, SVMs, and Random Forests) to best utilize their strengths and increase overall predictive accuracy.

2.3 Deep learning-based techniques

This study includes fully connected neural networks also known as dense networks, these are basic forms of deep neural networks where each neuron in one layer is connected to every neuron in the next layer.Fully connected neural networks are used to analyze complicated patterns retrieved from digitalized fine needle aspiration (FNA) images, improving breast cancer detection accuracy through sophisticated feature learning capabilities. While traditional methods have their strengths ,the significance of deep learning in the research work lies in its ability to enhance the accuracy, efficiency, and automation of breast cancer detection processes, thus contributing to advancements in diagnostic precision, improving accuracy through feature learning, and enhancing overall diagnostic capabilities in breast cancer detection.

3 Methodology

3.1 Dataset

We conducted our research utilizing the Winconsin Diagnostic Breast Cancer dataset,which has 65% data of one class and 35% of another. Feature's of this dataset were extracted from a digitized image of a fine needle aspirate (FNA) of a breast mass. This has a class distribution of 357 benign and 212 malignant .They describe the properties of the cell nuclei in the image and includes ten real valued features computed for each cell nucleus.

3.2 Data augmentation

3.3 SMOTE

The Wisconsin dataset is noted for having imbalanced classes, with 65% data belonging to one class and 35% to another. In order to correct this imbalance, the synthetic minority oversampling method (SMOTE) is used. SMOTE generates synthetic samples of the minority class to balance the dataset, which improves machine learning model performance, particularly when one class is underrepresented. This balanced data helped in achieving more reliable and robust model evaluations.

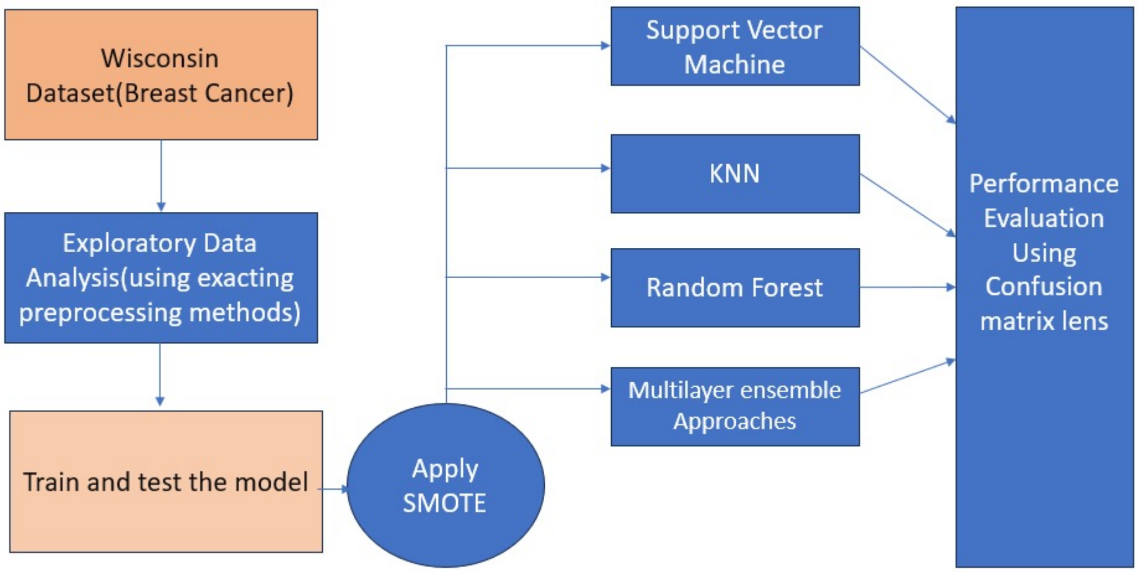


Fig 1: Framework of the Proposed architecture of Breast Cancer Detection

3.4 Method

We use different libraries like Scikit learn, imbalanced-learn and python programming language to conduct all the experiments discussed in this study , tensorflow was used to conduct experiments related to deep learning experiments. Then we perform exploratory data analysis using exacting preprocessing techniques .After exploring the data we perform preprocessing techniques where the original dataset comprises 33 features, but the first (id) and last (unnamed) columns add nothing to our outcome (whether is benign or malignant?), so they are removed from the dataset. A character feature of the dataset, "diagnostic," was manually turned into a numeric feature using a python dictionary where M=1 and B=0. Standardization uses StandardScalar().fit transform() for training and StandardScalar().transform() for testing.Standardization is conducted after splitting the train-test data to eliminate data leaks. We split the dataset into 30%test and 70%train after that we apply Smote alogrithm to the dataset. After initialising smote we fit smote to the train data. This provides us train data of smote , the accuracy obtained from the train data is 100%. This train smote data is later used in different machine learning algorithms like K-nearest neighbours, SVMs,Random Forests, Multilayer Ensemble Approaches, fully connected neural network. We evaluate the model with the help of confusion metrics lens and study the increase in the accuracy using smote train data.

- KNN:It is a supervised ML classification algorithm. This technique saves all of the training data and categorizes new data points based on how similar the nearest (with the shortest Euclidean distance) k training data points are. It's a non-parametric method, which means it doesn't make any predictions regarding the data. The algorithm does not instantly start learning from the training data provided; instead, it rests on the training data until new data for classification is provided. As a result, it's also known as a sluggish learning algorithm. The training data assumptions are nonparametric. It handles nonlinear data. Because KNN requires storing the training data, it can be time-consuming and space-consuming. In two-dimensional space, the Euclidean formula can be used to calculate the distance between two data points (x2, x1) and (y2, y1)

- SVM:It is another supervised ML technique for classification. This approach uses a hyperplane to partition the dataset into separate groups to ensure an optimum margin for accuracy. A hyperplane is a function that best separates classes, and a margin is the distance between the two nearest data points of distinct classes to the hyperplane. These models learn to categorize, forecast, and find outliers.
- Random Forest Classifer:RF classifier is a classification technique that uses supervised ML. This algorithm combines several DT classifiers on distinct subsets of the input dataset, as the name implies the algorithm increases its predictive accuracy by taking the average of their results to determine a single outcome. More trees improve accuracy and reduce the likelihood of overfitting. Ensemble learning is the name for this concept. The RF adds randomness to the model. Instead of the most important property, it randomly splits nodes. So, the model varies. RF is a technique for learning trees by aggregating bootstraps. Let X=x1, x2, x3, ..., xn) with replies Y=x1, x2, x3, ..., xn) with a lower limit of b=1 and an upper limit of B
- Multilayer Ensemble Approaches:In this study, multilayer ensemble approaches were employed to enhance breast cancer detection accuracy by combining multiple machine learning models. These approaches integrate various classifiers, such as K-nearest neighbors (KNN), support vector machines (SVMs), fully connected neural networks, and random forests. By leveraging the strengths of different algorithms, ensemble methods improve the robustness and generalization of predictions. The ensemble model demonstrated remarkable predictive power, efficiently identifying critical features and reducing false negatives. This method enhances overall diagnostic performance, offering a more reliable tool for early breast cancer detection.
- Fully Connected Neural Network:

3.5 Evaluation metrics

- The confusion matrix, accuracy, precision, recall, and other measures are used to assess how effective these algorithms are. Reduced false negatives, or missed detections, are especially emphasized in the context of breast cancer detection since early diagnosis and therapy depend on it.

- Confusion Matrix:The confusion matrix is a 2×2 matrix that comprises the letters true negative (TN), false positive (FP), false negative (FN), and true positive (TP) at locations 11, 12, 21, and 22, respectively. It divides the categorization model's expected outputs into four categories: TN (correctively rejected), FP (incorrectly accepted), FN (incorrectly rejected), and TP (correctly accepted).
- Accuracy:The proportion of all predictions that are predicted correctly (accurately) is referred to as accuracy. It is the measurement of properly categorized subjects relative to the total number of subjects.

4 Results

4.1 Qualitative results

Using SMOTE significantly improved the k-NN classifier's robustness by creating a more balanced training set. This led to better generalization capabilities, as evidenced by the increased accuracy.SMOTE effectively addressed class imbalance, which was previously skewed towards the majority class. This improvement was reflected in a more balanced confusion matrix with reduced false positives and false negatives, indicating better performance on the minority class,it contributed to a more comprehensive enhancement of the model's performance.

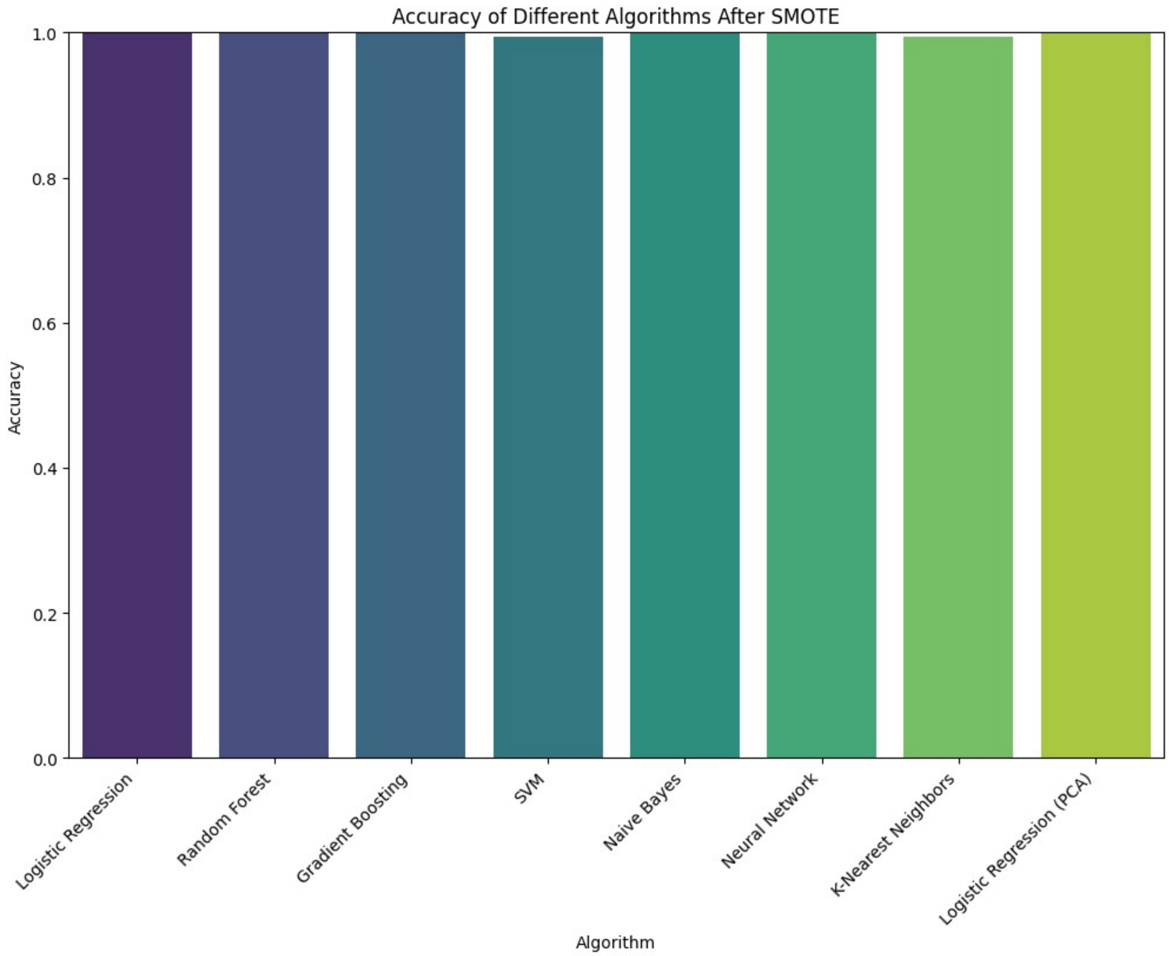
4.2 Quantitative results

After applying the smote algorithm we obtained a train smote data , the accuracy for the KNN model increased from 96.49% to 99%, for SVM it increased from 99.12 to 100%,for fully connected neural network accuracy increased to 100%.

Hence using the train smote data in different algorithm we saw an increase in the accuracy level of different algorithms.

5 Conclusion

Breast cancer is a prevalent and serious disease that affects millions of women worldwide. Early detection is crucial as it significantly improves survival rates and patient


Accuracy of Different Algorithms After SMOTE
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
Logistic Regression
Random Forest
Gradient Boosting
SVM
Naive Bayes
Neural Network
K-Nearest Neighbors
Logistic Regression (PCA)
Algorithm

outcomes. In recent years, advanced machine learning techniques have emerged as powerful tools for enhancing the accuracy and efficiency of breast cancer detection. This study explores the application of sophisticated machine and deep learning methods to improve the diagnostic process by leveraging digitalized fine needle aspirate (FNA) images in conjunction with the Multisurface Method-Tree (MSM-T) classification technique. To overcome the imbalance in WDBC dataset, the study employs the Synthetic Minority Oversampling Technique (SMOTE) to augment the training set. The research critically evaluates various machine learning algorithms, including K-Nearest Neighbors (KNN), Support Vector Machines (SVMs), Fully Connected Neural Networks, and Random Forests, to assess their effectiveness in detecting breast cancer patterns. These techniques are assessed individually and combined through multilayer ensemble methods to leverage their collective strengths. By examining the confusion matrix across three different folds of cross-validation, the study demonstrates the model's effectiveness through metrics such as lower false negative rates and higher recall rates.

The study's results highlight the model's remarkable predictive power and its ability to identify significant features relevant to breast cancer detection.Through this thorough investigation, the study aims to enhance the diagnostic toolkit available for breast cancer detection, striving to optimize diagnostic precision and ultimately improve patient outcomes.

References, if any. Remove this if there are no references.